

Exercise -I

Q1. Mean of first n natural numbers is-

- (A) $\frac{n(n-1)}{2}$ (B) $\frac{n(n+1)}{2}$ (C) $\frac{(n+1)}{2}$ (D) $\frac{n+1}{2}$

Ans. D

Sol. -

Q2. Following table shows the weight of 12 students:

Weight (in kgs.)	67	70	72	73	75
No. of students	4	3	2	2	1

then mean weight is-

- (A) 70.25 kg. (B) 70.50 kg. (C) 70.75 kg. (D) None of these

Ans. A

Sol. -

Q3. A factory employs 100 workers of whom 60 work in the first shift and 40 work in the second shift. The average wage of all the 100 workers is Rs.38. If the average wage of 60 workers of the first shift is Rs.40, then the average wage of the remaining 40 workers of the second shift is-

- (A) 35 (B) 40 (C) 45 (D) None of these

Ans. A

Sol. -

Q4. If $\bar{x}$ is the mean of a set of n observations $x_1, x_2, x_3, \dots, x_n$ then $\sum_{i=1}^n (x_i - \bar{x})$ is equal to

- (A) M.D. about mean (B) S.D. (C) 0 (D) None of these

Ans. C

Sol. -

Q5. If the mean of 3, 4, x, 7, 10 is 6, then the value of x is-

- (A) 4 (B) 5 (C) 6 (D) 7

Ans. C

Sol. -

Q6. Weighted mean is computed by the formula

- (A) $\frac{\sum x_i}{\sum w_i}$ (B) $\frac{\sum w_i}{\sum x_i}$ (C) $\frac{\sum x_i w_i}{\sum w_i}$ (D) $\frac{\sum w_i x_i}{\sum w_i}$

Ans. C

Sol. -

Q7. The mean of a set of numbers is $\bar{x}$. If each number is increased by α , the mean of the new set is-

- (A) $\bar{x}$ (B) $\bar{x} + \lambda$ (C) $\lambda \bar{x}$ (D) None of these

Ans. B

Sol. -

Q8. The mean of a set of numbers is $\bar{x}$. If such number is multiplied by λ , then the mean of the new set is-

- (A) $\bar{x}$ (B) $\lambda + \bar{x}$ (C) $\lambda \bar{x}$ (D) None of these

Ans. C

Sol. -

Q9. The mean of the squares of first n natural numbers is-

- (A) $\frac{1}{2}n^2$ (B) $\frac{1}{8}n(n+1)$ (C) $\frac{1}{6}n(2n+1)$ (D) $\frac{1}{6}(n+1)(2n+1)$

Ans. D

Sol. -

Q10. For a continuous series the mean is computed by the following formula-

- (A) $\text{Mean} = A + \frac{\sum f}{n}$ (B) $\text{Mean} = A + \frac{\sum d}{\sum f}$ (C) $\text{Mean} = A + \frac{\sum f}{\sum d}$
(D) $\text{Mean} = A + \frac{\sum fd}{\sum f}$

Ans. D

Sol. -

Q11. If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to-

- (A) 10 (B) 11 (C) 12 (D) None of these

Ans. B

Sol. -

Q12. The mean of discrete observations $y_1, y_2, \dots, y_n$ is given by-

$$(A) \frac{\sum_{i=1}^n y_i}{n} \quad (B) \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n i} \quad (C) \frac{\sum_{i=1}^n y_i f_i}{n} \quad (D) \frac{\sum_{i=1}^n y_i f_i}{\sum_{i=1}^n f_i}$$

Ans. A

Sol. -

Q13. If the mean of numbers 27, 31, 89, 107, 156 is 82, then the mean of 130, 126, 68, 50, 1 is-

(A) 75 (B) 157 (C) 82 (D) 80

Ans. A

Sol. -

Q14. If d_i is the deviation of a class mark y_i from 'a', the 'assumed mean' and f_i is the frequency, then

$Mg = x + \frac{1}{\sum f_i} (\sum f_i d_i)$, then x is-

(A) lower limit (B) assumed mean (C) number of observations (D) class size

Ans. B

Sol. -

Q15. The mean of first three terms is 14 and mean of next two terms is 18. The mean of all the five terms is-

(A) 14.5 (B) 15.0 (C) 15.2 (D) 15.6

Ans. D

Sol. -

Q16. Geometric mean of 3, 9 and 27 is-

(A) 18 (B) 6 (C) 9 (D) None of these

Ans. C

Sol. -

Q17. Geometric mean of the numbers $2, 2^2, 2^3, \dots, 2^n$ is

(A) $2^{2/n}$ (B) $2^{n/2}$ (C) $2^{\frac{n-1}{2}}$ (D) $2^{\frac{n+1}{2}}$

Ans. D

Sol. -

Q18. The geometric mean of n observations $x_1, x_2, x_3, \dots, x_n$ is

- (A) $\frac{\sum_{i=1}^n x_i}{n}$ (B) $\frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i}\right)}$ (C) $(x_1 x_2 x_3 \dots x_n)^{1/n}$ (D) None of these

Ans. C

Sol. -

Q19. The geometric mean of the numbers 4, 8, 16 is-

- (A) $\frac{28}{3}$ (B) 8 (C) $\frac{48}{7}$ (D) None

Ans. B

Sol. -

Q20. The harmonic mean of 4, 8, 16 is-

- (A) 6.4 (B) 6.7 (C) 6.85 (D) 7.8

Ans. C

Sol. -

Q21. The harmonic mean of 3, 7, 8, 10, 14 is-

- (A) $\frac{3+7+8+10+14}{5}$ (B) $\frac{1}{3} + \frac{1}{7} + \frac{1}{8} + \frac{1}{10} + \frac{1}{14}$ (C) $\frac{\frac{1}{3} + \frac{1}{7} + \frac{1}{8} + \frac{1}{10} + \frac{1}{14}}{5}$
 (D) $\frac{5}{\frac{1}{3} + \frac{1}{7} + \frac{1}{8} + \frac{1}{10} + \frac{1}{14}}$

Ans. D

Sol. -

Q22. In an arranged series of n observations (n being an odd number), the median is the value of -

- (A) $\left(\frac{n}{2}\right)$ th term (B) $\left(\frac{n+1}{2}\right)$ th term (C) $\left(\frac{n}{2} + 1\right)$ th term (D) $\left(n + \frac{1}{2}\right)$ th term

Ans. B

Sol. -

Q23. The median of 10, 14, 11, 9, 8, 12, 6 is

- (A) 10 (B) 12 (C) 14 (D) 11

Ans. A

Sol. -

Q24. If a variable takes the discrete values $\alpha + 4, \alpha - \frac{7}{2}, \alpha - \frac{5}{2}, \alpha - 3, \alpha - 2, \alpha + \frac{1}{2}, \alpha - \frac{1}{2}, \alpha + 5$ ($\alpha > 0$), then the median is :

(A) $\alpha - \frac{5}{4}$ (B) $\alpha - \frac{1}{2}$ (C) $\alpha - 2$ (D) $\alpha + \frac{5}{4}$

Ans. A

Sol. Arranged series

$$\alpha - \frac{7}{2}, \alpha - 3, \alpha - \frac{5}{2}, \alpha - 2, \alpha - \frac{1}{2}, \alpha + \frac{1}{2}, \alpha + 4, \alpha + 5$$

$$\begin{aligned} \text{Median} &= \frac{\left(\frac{8}{2}\right)^{\text{th}} \text{term} + \left(\frac{8}{2} + 1\right)^{\text{th}} \text{term}}{2} \\ &= \frac{4^{\text{th}} \text{term} + 5^{\text{th}} \text{term}}{2} \Rightarrow \frac{2\alpha - \frac{5}{2}}{2} \\ &= \frac{\alpha - 2 + \alpha - \frac{1}{2}}{2} = \alpha - \frac{5}{4} \end{aligned}$$

Q25. In an arranged discrete series in which total number of observations 'n' is even, median is

- (A) $\frac{n}{2}$ th term (B) $\left(\frac{n}{2} + 1\right)$ th term (C) the mean of $\frac{n}{2}$ th and $\left(\frac{n}{2} + 1\right)$ th term
(D) None of these

Ans. C

Sol. -

Q26. The median of the items 6, 10, 4, 3, 9, 11, 22, 18 is-

- (A) 9 (B) 10 (C) 9.5 (D) 11

Ans. C

Sol. -

Q27. The mode of the following items is 0, 1, 6, 7, 2, 3, 7, 6, 6, 2, 6, 0, 5, 6, 0 is-

- (A) 0 (B) 5 (C) 6 (D) 2

Ans. C

Sol. -

Q28. Mode of the distribution is

Marks	4	5	6	7	8
Number of students	3	5	10	6	1

- (A) 6 (B) 10 (C) 8 (D) None of these

Ans. A

Sol. -

Q29. For a normal distribution, we have

- (A) mean = median (B) median = mode (C) mode = mean
(D) mean = median = mode

Ans. D

Sol. -

Q30. If median = (mode + 2 mean) / 3, then M is equal to-

- (A) 3 (B) 1/3 (C) 2 (D) None of these

Ans. B

Sol. -

Q31. The relationship between mean, median and mode for a moderately skewed distribution is-

- (A) mode = median – 2 mean (B) mode = 2 median – mean
(C) mode = 2 median – 3 mean (D) mode = 3 median – 2 mean

Ans. D

Sol. -

Q32. If the mode of a data is 18 and the mean is 24, then median is-

- (A) 18 (B) 24 (C) 22 (D) 21

Ans. C

Sol. -

Q33. The scores of a batsman in ten innings are: 38, 70, 48, 34, 42, 55, 63, 46, 54, 44, then mean deviation about the median is:

- (A) 8.4 (B) 8.5 (C) 8.6 (D) 8.8

Ans. C

Sol. -

Q34. If the Standard Deviation of a variate is s , then the S.D. of $3x + 2$ is:

- (A) σ (B) 3σ (C) $3\sigma + 2$ (D) None of these

Ans. B

Sol. -

Q35. For a frequency distribution, the mean deviation about mean is computed by

$$(A) \text{M.D.} = \frac{\sum d_i}{\sum f_i} \quad (B) \text{M.D.} = \frac{\sum f_i d_i}{\sum f_i} \quad (C) \text{M.D.} = \frac{\sum f_i |d_i|}{\sum f_i} \quad (D) \text{M.D.} = \frac{\sum f_i}{\sum f_i |d_i|}$$

Ans. C

Sol. -

Q36. Mean deviation from the mean for the observations $-1, 0, 4$ is

$$(A) \sqrt{\frac{14}{3}} \quad (B) \frac{2}{3} \quad (C) 2 \quad (D) \text{None of these}$$

Ans. C

Sol. -

Q37. The variance of 2, 4, 6, 8, 10 is

$$(A) 8 \quad (B) \sqrt{8} \quad (C) 6 \quad (D) \text{None of these}$$

Ans. A

Sol. -

Q38. The S.D. of 7 scores 1, 2, 3, 4, 5, 6, 7 is

$$(A) 4 \quad (B) 2 \quad (C) \sqrt{7} \quad (D) \text{None of these}$$

Ans. B

Sol. -

Q39. If the standard deviation of 1, 2, 3, 4,..., 10 is σ , then the standard deviation of 11, 12, 13, 14,..., 20 is

$$(A) \sigma + 10 \quad (B) 10 \sigma \quad (C) \sigma \quad (D) \text{None of these}$$

Ans. C

Sol. -

Q40. S.D. of n observations $a_1, a_2, a_3, \dots, a_n$ is σ , then the S.D. of the observations $\lambda a_1, \lambda a_2, \lambda a_3, \lambda a_n$ is-

$$(A) \lambda \sigma \quad (B) -\lambda \sigma \quad (C) |\lambda| \sigma \quad (D) \sigma$$

Ans. C

Sol. -

Q41. If each observation of a raw data whose variance is σ^2 , is increased by λ , then the variance of the new set is-

$$(A) \sigma^2 \quad (B) \lambda^2 \sigma^2 \quad (C) \lambda + \sigma^2 \quad (D) \lambda^2 + \sigma^2$$

Ans. A

Sol. -

Q42. Let σ be the standard deviation of n observations. Each of the n observations is multiplied by a constant c . Then the standard deviation of the resulting numbers is :

- (A) σ (B) $|c| \cdot \sigma$ (C) $\sigma\sqrt{c}$ (D) None of these

Ans. B

Sol.
$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

$$\sigma_1 = \sqrt{\frac{\sum (cx_i - c\bar{x})^2}{N}}$$

$$\sigma_1 = |c| \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

$$\sigma_1 = |c|\sigma$$

Q43. For a frequency distribution standard deviation is computed by-

(A) $\sigma = \frac{\sum f(x - \bar{x})}{\sum f}$ (B) $\sigma = \frac{\sqrt{\sum f(x - \bar{x})^2}}{\sum f}$ (C) $\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}}$

(D) $\sigma = \sqrt{\frac{\sum f(x - \bar{x})}{\sum f}}$

Ans. C

Sol. -

Exercise -II

Q1. If the mean of the first n odd natural numbers be n itself, then n is-

- (A) 1 (B) 2 (C) 3 (D) any natural number

Ans. D

Sol. -

Q2. The A.M. of the series 1, 2, 4, 8, 16, ..., 2^n is-

- (A) $\frac{2^n - 1}{n}$ (B) $\frac{2^{n+1} - 1}{n + 1}$ (C) $\frac{2^n + 1}{n}$ (D) $\frac{2^n - 1}{n + 1}$

Ans. B

Sol. -

Q3. If the mean of the set of numbers $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$, then the mean of the numbers $x_i + 2, 1 \leq i \leq n$, is-

- (A) $\bar{x} + 2n$ (B) $\bar{x} + n + 1$ (C) $\bar{x} + 2$ (D) $\bar{x} + n$

Ans. B

Sol. -

Q4. Mean of the first n terms of the A.P. $a + (a + d) + (a + 2d) + \dots$ is

- (A) $a + \frac{nd}{2}$ (B) $a + \frac{(n-1)d}{2}$ (C) $a + (n-1)d$ (D) $a + nd$

Ans. B

Sol. -

Q5. If μ is the mean of a distribution, then $\sum f_i(y_i - \mu)$ is equal to

- (A) M.D. (B) S.D. (C) 0 (D) None of these

Ans. C

Sol. -

Q6. If the mean of n observations $1^2, 2^2, 3^2, \dots, n^2$ is $\frac{46n}{11}$, then n is equal to

- (A) 11 (B) 12 (C) 23 (D) 22

Ans. A

Sol. -

Q7. The mean of 50 observations is 36. If two observations 30 and 42 are deleted, then the mean of the remaining observations is-

- (A) 48 (B) 36 (C) 38 (D) None of these

Ans. D

Sol. -

Q8. The weighted mean of first n natural numbers, whose weights are equal to the squares of corresponding numbers, is

- (A) $\frac{n+1}{2}$ (B) $\frac{3n(n+1)}{2(2n+1)}$ (C) $\frac{(n+1)}{2n+1}$ (D) $\frac{n(n+1)}{2}$

Ans. B

Sol. -

Q9. A group of 10 items has mean 6. If the mean of 4 of these items is 7.5, then the mean of the remaining items is-

- (A) 6.5 (B) 5.5 (C) 4.5 (D) 5.0

Ans. D

Sol. -

Q10. The mean of a set of observations is $\bar{x}$. If each observation is divided by α , $\alpha \neq 0$, and then is increased by 10 then the mean of the new set is-

- (A) $\frac{\bar{x}}{\alpha}$ (B) $\frac{\bar{x} + 10}{\alpha}$ (C) $\frac{\bar{x} + 10\alpha}{\alpha}$ (D) $a\bar{x} + 10$

Ans. C

Sol. $\frac{x_1 + x_2 + \dots + x_n}{n} = \bar{x}$

if each observation is divided by α ($\alpha \neq 0$) and then increased by 10 then the new mean of the new set

$$\frac{\left(\frac{x_1}{\alpha} + 10\right) + \left(\frac{x_2}{\alpha} + 10\right) + \dots + \left(\frac{x_n}{\alpha} + 10\right)}{n}$$

$$\Rightarrow \frac{(x_1 + x_2 + \dots + x_n) + 10\alpha n}{\alpha n} = \frac{\bar{x} + 10\alpha}{\alpha}$$

Q11. The geometric mean of the first n terms of the G.P. $a + ar + ar^2 + \dots$ is-

- (A) $ar^{n/2}$ (B) ar^n (C) $ar^{(n-1)/2}$ (D) ar^{n-1}

Ans. C

Sol. -

Q12. The geometric mean of the observations 2, 4, 8, 16, 32, 64 is-

- (A) $2^{5/2}$ (B) $2^{7/2}$ (C) 33 (D) None of these

Ans. B

Sol. -

Q13. A boy goes to school from his home at a speed of x km/hr. and comes back at a speed of y km/hr. then the average speed of the boy is given by-

- (A) $\frac{x+y}{2}$ km/hr. (B) $\sqrt{xy}$ km/hr. (C) $\frac{2xy}{x+y}$ km/hr. (D) Any of these

Ans. C

Sol. -

Q14. Ram spends equal amounts on purchasing three kinds of pens being sold at Rs.5, Rs.10, Rs.15 per piece. Average cost of each pen is-

- (A) Rs.10 (B) Rs. $\frac{90}{11}$ (C) Rs.9 (D) None of these

Ans. B

Sol. -

Q15. An automobile driver travels from plain to a hill station 120km distant at an average speed of 30 km.per hour. He then makes the return trip at an average speed of 25 km. per hour. He covers another 120 km. distance on plane at an average speed of 50 km. per hour. His average speed over the entire distance of 360 km. will be-

- (A) $\frac{30+25+50}{3}$ km/hr. (B) $(30 \times 25 \times 50)^{1/3}$ (C) $\frac{3}{\frac{1}{30} + \frac{1}{25} + \frac{1}{50}}$ km/hr.

(D) None of these

Ans. C

Sol. $T_1 = \frac{120}{30}$

$$T_2 = \frac{120}{25}$$

$$T_3 = \frac{120}{50}$$

$$\begin{aligned} \text{Avg. speed} &= \frac{\text{Total distance}}{\text{Time Taken}} = \frac{3 \times 120}{T_1 + T_2 + T_3} \\ &= \frac{3}{\frac{1}{30} + \frac{1}{25} + \frac{1}{50}} \text{ km/hr} \end{aligned}$$

Q16. If a, b, c are any three positive numbers, then the least value of $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$ is-

(A) 3 (B) 6 (C) 9 (D) None of these

Ans. C

Sol. -

Q17. Product of n positive numbers is unity. The sum of these numbers can not be less than-

(A) 1 (B) n (C) n^2 (D) None of these

Ans. B

Sol. -

Q18. The median of the data 13, 14, 16, 18, 20, 22 is-

(A) 17 (B) 16 (C) 18 (D) None of these

Ans. A

Sol. -

Q19. For a continuous series, the mode is computed by the formula

(A) $\ell + \frac{f_{m-1}}{f_m - f_{m-1} - f_{m+1}} \times c$ (B) $\ell + \frac{f_m - f_{m-1}}{f_m - f_{m-1} - f_{m+1}} \times c$ (C) $\ell + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times c$
(D) $\ell + \frac{2f_m - f_{m-1}}{f_m - f_{m-1} - f_{m-2}} \times c$

Ans. C

Sol. -

Q20. If mean = (3 median – mode) x, then the value of x is-

(A) 1 (B) 2 (C) 1/2 (D) 3/2

Ans. C

Sol. -

Q21. Which of the following is not a measure of dispersion?

(A) variance (B) mean deviation (C) standard deviation (D) mode

Ans. D

Sol. -

Q22. Which of the following is not a measure of dispersion ?

(A) Mean (B) Variance (C) Mean deviation (D) Range

Ans. A

Sol. -

Q23. The standard deviation of a variate x is σ . The standard deviation of the variable $\frac{ax+b}{c}$; a, b, c are constants, is-

- (A) $\left(\frac{a}{c}\right)\sigma$ (B) $\left|\frac{a}{c}\right|\sigma$ (C) $\left(\frac{a^2}{c^2}\right)\sigma$ (D) None of these

Ans. B

Sol. $y = \frac{ax+b}{c} = \frac{a}{c}x + \frac{b}{c}$

$$\bar{y} = \frac{a}{c}\bar{x} + \frac{b}{c}$$

$$y - \bar{y} = \frac{a}{c}(x - \bar{x})$$

$$(y - \bar{y})^2 = \left(\frac{a}{c}\right)^2 (x - \bar{x})^2$$

$$\sum (y - \bar{y})^2 = \left(\frac{a}{c}\right)^2 \sum (x - \bar{x})^2$$

$$\sum \frac{(y - \bar{y})^2}{n} = \left(\frac{a}{c}\right)^2 \sum \frac{(x - \bar{x})^2}{n}$$

$$\sigma_y^2 = \left(\frac{a}{c}\right)^2 \sum \frac{(x - \bar{x})^2}{n}$$

$$\sigma_y^2 = \left(\frac{a}{c}\right)^2 \sigma_x^2$$

$$\sigma_y = \left|\frac{a}{c}\right| \sigma_x \{\sigma_x = \sigma\}$$

$$\sigma_y = \left|\frac{a}{c}\right| \sigma$$

Q24. The S.D. of 15 items is 6 and if each item is decreased by 1, then standard deviation will be-

- (A) 5 (B) 7 (C) $9\frac{1}{25}$ (D) 6

Ans. D

Sol. -

Q25. The S.D of the first n natural numbers is-

- (A) $\frac{n+1}{2}$ (B) $\sqrt{\frac{n(n+1)}{2}}$ (C) $\sqrt{\frac{n^2-1}{12}}$ (D) None of these

Ans. C

Sol. -

Exercise -III

Q1. If G_1, G_2 are the geometric means of two series of observations and G is the geometric mean of the ratios of the corresponding observations, then the value of G -

- (A) $\frac{G_1}{G_2}$ (B) $\frac{\log G_1}{\log G_2}$ (C) $\log(G_1.G_2)$ (D) $\log G_1 - \log G_2$

Ans. A

Sol. -

Q2. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set -

- (A) remains the same as that of the original set (B) is increased by 2
(C) is decreased by 2 (D) is two times the original median

Ans. A

Sol. Let $A_1, A_2, A_3, \dots, A_9$ are 9 observations in increasing order.
Median of these observations is 5th observation.

$$\Rightarrow A_5 = 20.5$$

Largest four observations are $= A_6, A_7, A_8, A_9$

after increasing by 2 $= A_6 + 2, A_7 + 2, A_8 + 2, A_9 + 2$

Hence, now observations

$A_1, A_2, A_3, \dots, A_5, A_6 + 2, A_7 + 2, A_8 + 2, A_9 + 2$

Median of new set is still A_5

$\Rightarrow$ Median of new set will remain the same as that is initial.

Q3. In an experiment with 15 observations on x , the following results were available :

$$\sum x^2 = 2830, \sum x = 170$$

One observation that was 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is

- (A) 8.33 (B) 78.00 (C) 188.66 (D) 177.33

Ans. B

Sol. -

Q4. Consider the following statements:

- (a) Mode can be computed from histogram
(b) Median is not independent of change of scale
(c) Variance is independent of change of origin and scale.
Which of these is/ are correct ?

(A) only (a) (B) only (b) (C) only (a) and (b) (D) (a), (b) and (c)

Ans. C

Sol. -

Q5. In a series of $2n$ observations, half of them equal a and remaining half equal $-a$. If the standard deviation of the observations is 2, then $|a|$ equals-

(A) $1/n$ (B) $\sqrt{2}$ (C) 2 (D) $\frac{\sqrt{2}}{n}$

Ans. C

Sol. -

Q6. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately

(A) 22.0 (B) 20.5 (C) 25.5 (D) 24.0

Ans. D

Sol. -

Q7. Let $x_1, x_2, \dots, x_n$ be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is -

(A) 15 (B) 18 (C) 9 (D) 12

Ans. B

Sol. -

Q8. Suppose a population A has 100 observations 101, 102, ..., 200, and another population B has 100 observations 151, 152, ..., 250. If V_A and V_B represent the variances of the two populations, respectively, then $\frac{V_A}{V_B}$ is-

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$ (C) 2 (D) 1

Ans. D

Sol. -

Q9. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is-

(A) 40 (B) 20 (C) 80 (D) 60

Ans. C

Sol. -

Q10. The mean of the numbers $a, b, 8, 5, 10$ is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b ?

- (A) $a = 5, b = 2$ (B) $a = 1, b = 6$ (C) $a = 3, b = 4$ (D) $a = 0, b = 7$

Ans. C

Sol. -

Q11. If the mean deviation of the numbers $1, 1 + d, 1 + 2d, \dots, 1 + 100d$ from their mean is 255, then the d is equal to -

- (A) 10.0 (B) 20.0 (C) 10.1 (D) 20.2

Ans. C

Sol. -

Q12.

Ans. -

Sol. -

Q13. For two data sets each of size is 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4 respectively, then the variance of the combined data set is :-

- (A) $\frac{5}{2}$ (B) $\frac{11}{2}$ (C) 6 (D) $\frac{13}{2}$

Ans. B

Sol. -

Q14. If the mean deviation about the median of the number $a, 2a, \dots, 50a$ is 50, then $|a|$ equal:

- (A) 4 (B) 5 (C) 2 (D) 3

Ans. A

Sol. -

Q15. A scientist is weighing each of 30 fishes. Their mean weight worked out is 30 gm and a standard deviation of 2 gm. Later, it was found that the measuring scale was misaligned and always under reported every fish weight by 2 gm. The correct mean and standard deviation (in gm) of fishes are respectively :

- (A) 28, 4 (B) 32, 2 (C) 32, 4 (D) 28, 2

Ans. B

Sol. -

Q16. Let $x_1, x_2, \dots, x_n$ be n observations, and let $\bar{x}$ be their arithmetic mean and σ^2 be their variance.

Statement -1: Variance of $2x_1, 2x_2, \dots, 2x_n$ is $4\sigma^2$.

Statement-2: Arithmetic mean of $2x_1, 2x_2, \dots, 2x_n$ is $4\bar{x}$.

(A) Statement-1 is true, Statement-2 is false.

(B) Statement-2 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(D) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

Ans. 5f433b410f9f218bad390d3c

Sol. -

Q17. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given ?

(A) mean (B) median (C) mode (D) variance

Ans. D

Sol. -

Q18. The variance of first 50 even natural numbers is

(A) 437 (B) $\frac{437}{4}$ (C) $\frac{833}{4}$ (D) 833

Ans. D

Sol. -

Q19. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3, 4 and 5 and added to the data, then the mean of the resultant data, is :

(A) 16.8 (B) 16.0 (C) 15.8 (D) 14.0

Ans. D

Sol. -

Q20. If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ?

(A) $3a^2 - 34a + 91 = 0$ (B) $3a^2 - 23a + 44 = 0$ (C) $3a^2 - 26a + 55 = 0$

$$(D) 3a^2 - 32a + 84 = 0$$

Ans. D

$$\text{Sol. } \sigma^2 = (3.5)^2 = 12.25$$

$$\sigma^2 = \frac{1}{n} \sum x^2 - (\bar{x})^2$$

$$12.25 = \frac{1}{4}(2^2 + 3^2 + a^2 + 11^2) - \left(\frac{2+3+a+11}{4}\right)^2$$

$$3a^2 - 32a + 84 = 0$$

Q21. If $\sum_{i=1}^9 (x_i - 5)$ and $\sum_{i=1}^9 (x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, \dots, x_9$ is

(A) 2 (B) 3 (C) 9 (D) 4

Ans. A

Sol. -

Q22. The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to :

(A) -20 (B) -10 (C) -5 (D) 10

Ans. A

Sol. New mean $\bar{x}_1 = p\bar{x} - q$

$$10 = p(20) - q \quad \dots\dots\dots(1)$$

and New s.d.

$$\sigma_2 = |\sigma| \sigma_1$$

$$1 = |p| \sigma_1$$

$$p = \pm \frac{1}{2}$$

$$\text{If } p = \frac{1}{2}$$

then $q = 0$ from equation (1)

$$\text{If } p = -\frac{1}{2}$$

then $q = -20$

Q23. The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11, then the **correct** variance is

(A) 3.99 (B) 4.01 (C) 4.02 (D) 3.98

Ans. A

Sol. $\frac{\sum x_i}{20} = 10 \dots (i) \frac{\sum x_i^2}{20} - 100 = 4 \dots (ii) \sum x_i^2 = 104 \times 20 = 2080$

$$\text{Actual mean} = \frac{200 - 9 + 11}{20} = \frac{202}{20}$$

$$\begin{aligned} \text{Variance} &= \frac{2080 - 81 + 121}{20} - \left(\frac{202}{20}\right)^2 \\ &= \frac{2120}{20} - (10.1)^2 = 106 - 102.01 = 3.99 \end{aligned}$$

Q24. Let the observations $x_i (1 \leq i \leq 10)$ satisfy the equations, $\sum_{i=1}^{10} (x_i - 5) = 10$ and

$\sum_{i=1}^{10} (x_i - 5)^2 = 40$. If μ and λ are the mean and the variance of the observations,

$x_1 - 3, x_2 - 3, \dots, x_{10} - 3$, then the ordered pair (μ, λ) is equal to :

(A) (6, 3) (B) (6, 6) (C) (3, 6) (D) (3, 3)

Ans. D

Sol. $\sum_{i=1}^{10} (x_i - 5) = 10 \dots (1)$

$\sum_{i=1}^{10} (x_i - 5)^2 = 40 \dots (2)$

$$\text{mean } (x_i - 5) = \frac{\sum (x_i - 5)}{10} = \frac{10}{10} = 1$$

$$\mu = \text{mean } (x_i - 3) = \text{mean } (x_i - 5 + 2) = 1 + 2 = 3$$

$$\lambda = \text{var}(x_i - 3) = \text{var}(x_i - 5) = \frac{\sum (x_i - 5)^2}{10} - \frac{\sum (x_i - 5)}{10}$$

$$= \frac{40}{10} - 1$$

$$= 4 - 1 = 3$$

Q25.

Ans. -

Sol. -

Q26.

Ans. -

Sol. -

Q27. Let $X = \{x \in \mathbb{N} : 1 \leq x \leq 17\}$ and $Y = \{ax + b : x \in X \text{ and } a, b \in \mathbb{R}, a > 0\}$. If mean and variance of elements of Y are 17 and 216 respectively then $a + b$ is equal to :

(A) 9 (B) -7 (C) 7 (D) -27

Ans. B

Sol. $X = \{1, 2, \dots, 17\}$
Mean = $\frac{1 + 2 + \dots + 17}{17} = 9$
Variance = $\frac{\sum_{i=1}^{17} x_i^2}{17} - (\bar{x})^2 = 105 - 81 = 24$
 $Y = \{ax + b : x \in X \text{ and } a, b \in \mathbb{R}, a > 0\}$
Mean = $9a + b = 17$
Variance = $24a^2 = 216$
 $\Rightarrow a^2 = 9 \Rightarrow a = 3$
 $b = -10$

Q28.

Ans. -

Sol. -

Q29. For the frequency distribution :

Variate (x) : $x_1 \quad x_2 \quad x_3 \dots x_{15}$

Frequency (f) : $f_1 \quad f_2 \quad f_3 \dots f_{15}$

where $0 < x_1 < x_2 < x_3 < \dots < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, the standard deviation cannot be :

(A) 6 (B) 1 (C) 2 (D) 4

Ans. A

Sol. standard deviation will be maximum when half x_i approach to 0 are half to 10.

$\bar{x}$ will be middle value = 5

$$(x_i - \bar{x})^2 \leq 25$$

$$(x_i - \bar{x}) \leq 5; \sigma \leq 5$$

Q30. Let $x_i (1 \leq i \leq 10)$ be ten observation of a random variable X. If $\sum_{i=1}^{10} (x_i - p) = 3$ and

$\sum_{i=1}^{10} (x_i - p)^2 = 9$ where $0 \neq p \in \mathbb{R}$, then the standard deviation of these observation is :

(A) $\sqrt{\frac{3}{5}}$ (B) $\frac{4}{5}$ (C) $\frac{9}{10}$ (D) $\frac{7}{10}$

Ans. C

Sol. $\sum_{i=1}^{10} (x_i - p) = 3$

$$\sum_{i=1}^{10} (x_i - p)^2 = 9$$

If we add or subtract a particular number in all data then S.D remains unchanged.

Put $X_i = x_i - p$;

S.D. of X_i and x_i are equal.

$$\text{hence } \sum_{i=1}^{10} X_i = 3$$

$$\sum_{i=1}^{10} X_i^2 = 9$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{10} X_i^2 - (\bar{X})^2$$

$$\sigma^2 = \frac{1}{10}(9) - \left(\frac{3}{10}\right)^2$$

$$\sigma^2 = \frac{90 - 9}{100}$$

$$\text{S.D.} = \sigma = \frac{9}{10}$$

Q31. The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5, 7, 10, 12, 14, 15, then the absolute difference of the remaining two observations is :

(A) 5 (B) 9 (C) 3 (D) 7

Ans. D

Sol. Let remaining two observations are a and b.

$$\text{Mean} = 10 = \frac{5 + 7 + 12 + 10 + 15 + 14 + a + b}{8}$$

$$\Rightarrow 80 = 63 + a + b$$

$$a + b = 17 \quad \dots\dots\dots\text{(I)}$$

$$\text{Variance} = \sigma^2 = \frac{1}{n} \sum x_i^2 - (\bar{x})^2$$

$$\Rightarrow 13.5 = \frac{1}{2} (25 + 49 + 144 + 100 + 225 + 196 + a^2 + b^2) - 100$$

$$\Rightarrow 108 = 739 + a^2 + b^2 - 800$$

$$\Rightarrow 908 = 739 + a^2 + b^2$$

$$\Rightarrow a^2 + b^2 = 169$$

$$\Rightarrow (a + b)^2 - 2ab = 169$$

$$\Rightarrow 289 - 169 = 2ab \quad \quad \quad [\text{use (I)}]$$

$$ab = 60$$

$$(a - b)^2 = (a + b)^2 - 4ab$$

$$\Rightarrow (a - b)^2 = 289 - 240$$

$$\Rightarrow (a - b)^2 = 49$$

$$\therefore |a - b| = 7$$

Q32. If the variance of the following frequency distribution :

Class	:	10 – 20	20 – 30	30 – 40
Frequency	:	2	x	2

is 50, then x is equal to ____.

Ans. 4

Sol.

x_i	5	15	25
f_i	2	x	2

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{10 + 15x + 50}{4 + x}$$

$$= \frac{15(4 + x)}{4 + x} = 15$$

$$\sigma^2 = 50 = \frac{\sum f_i x_i^2}{\sum f_i} - (\bar{x})^2 = 50$$

$$\frac{50 + 225x + 1250}{4 + x} - (15)^2 = 50$$

$$x = 4$$

Q33. The mean and variance of 7 observations are 8 and 16, respectively. If five observations are 2, 4, 10, 12, 14, then the absolute difference of the remaining two observations is :

(A) 4 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Let the remaining observations are x and y.

$$\text{Mean} = \frac{2 + 4 + 10 + 12 + 14 + x + y}{7} = 8$$

$$\Rightarrow 42 + x + y = 56 \Rightarrow x + y = 14 \dots\dots\dots (1)$$

$$\text{Variance} = \frac{\sum_{i=1}^7 x_i^2}{7} - (\bar{x})^2 = 16 \Rightarrow \frac{2^2 + 4^2 + 10^2 + 12^2 + 14^2 + x^2 + y^2}{7} - (8)^2 = 16$$

$$\Rightarrow 460 + x^2 + y^2 = 560 \Rightarrow x^2 + y^2 = 100 \dots\dots\dots (2)$$

$$(1)^2 - 2 \dots\dots \Rightarrow 2xy = 196 - 100 = 96$$

$$\Rightarrow xy = 48 \dots\dots\dots (3)$$

From (1) and (3)

$$x + \frac{48}{x} = 14 \Rightarrow x^2 - 14x + 48 = 0$$

$$\Rightarrow x = 6 \text{ and } y = 8 \text{ or } x = 8 \text{ and } y = 6$$

$$\Rightarrow |x - y| = 2$$

Q34. If the mean and the standard deviation of the data 3, 5, 7, a, b are 5 and 2 respectively, then a and b are the roots of the equation :

$$(A) x^2 - 10x + 19 = 0 \quad (B) 2x^2 - 20x + 19 = 0 \quad (C) x^2 - 20x + 18 = 0$$

$$(D) x^2 - 10x + 18 = 0$$

Ans. A

Sol. Mean = $\frac{5+3+7+a+b}{5} = 5a+b=10$ $S.D = \left[\frac{5^2+3^2+7^2+a^2+b^2}{5} - 5^2 \right]^{\frac{1}{2}} = 2 \frac{83+(a+b)^2-2ab}{5} - 25 =$
 $83 + (10)^2 - 2ab - 125 = 20ab = 19$
 $equation x^2 - (a+b)x + ab = 0 x^2 - (10)x + 19 = 0$

Q35. If $\sum_{i=1}^n (x_i - a) = n$ and $\sum_{i=1}^n (x_i - a)^2 = na$, ($n, a > 1$) then the standard deviation of n observation $x_1, x_2, \dots, x_n$ is :

(A) $\sqrt{n(a-1)}$ (B) $n\sqrt{a-1}$ (C) $a-1$ (D) $\sqrt{a-1}$

Ans. D

Sol. $\sigma^2 = \frac{\sum (x_i - a)^2}{n} - \left(\frac{\sum (x_i - a)}{n} \right)^2 = \frac{na}{n} - \left(\frac{n}{n} \right)^2 = a - 1$
 $\sigma = \sqrt{a-1}$

Q36. Consider the data on x taking the values $0, 2, 4, 8, \dots, 2^n$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to

_____.

Ans. 6

Sol.

x	0	2	4	8		2^n
f	nC_0	nC_1	nC_2	nC_3		nC_n

$$\text{Mean} = \frac{\sum x_i f_i}{\sum f_i} = \frac{\sum_{r=0}^n 2^r {}^nC_r}{\sum_{r=0}^n {}^nC_r}$$

$$\text{Mean} = \frac{3^n - {}^nC_0}{2^n} = \frac{728}{2^n} \Rightarrow 3^n - 1 = 728 \Rightarrow 3^n = 729 \Rightarrow n = 6$$

Q37. If the variance of 10 natural numbers $1, 1, 1, \dots, 1, k$ is less than 10, then the maximum possible value of k is _____.

Ans. 11

Sol. $\sigma^2 < 10$
 $\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 < 10$
 $\frac{9+k^2}{10} - \left(\frac{9+k}{10} \right)^2 < 10$
 $9(k^2 - k) < 991$
 $k(k-2) < \frac{991}{9}$

$k = 11$ is largest integral value satisfying given inequality.

Q38. Let $X_1, X_2, \dots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18} (X_i - \alpha) = 36$ and $\sum_{i=1}^{18} (X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of

these observations is 1, then the value of $|\alpha - \beta|$ is _____.

Ans. 4

Sol.
$$\sum_{i=1}^{18} (x_i - \alpha) = 36$$
$$\therefore \sum_{i=1}^{18} x_i - 18\alpha = 36$$
$$\sum_{i=1}^{18} x_i = 36 + 18\alpha$$
$$\sum_{i=1}^{18} (x_i - \beta) = 90$$
$$\sum_{i=1}^{18} x_i^2 - 2\beta \sum_{i=1}^{18} x_i + 18\beta^2 = 90$$
$$\sum_{i=1}^{18} x_i^2 = 90 + 2\beta(36 + 18\alpha) - 18\beta^2$$

Variance of these observations

$$6^2 = \frac{1}{18} \sum_{i=1}^{18} x_i^2 - \left(\frac{\sum x_i}{18} \right)^2$$
$$\Rightarrow 1 = \frac{1}{18} (90 + 3\beta(36 + 18\alpha) - 18\beta^2) - (2 + \alpha)^2$$
$$\Rightarrow 1 = 5 + 2\beta(2 + \alpha) - \beta^2 - (\alpha + 2)^2$$
$$\Rightarrow (\alpha + 2)^2 + \beta^2 - 2\beta(\alpha + 2) = 4$$
$$\Rightarrow (\alpha + 2 - \beta)^2 = 4$$
$$\Rightarrow (\alpha - \beta + 2)^2 - 4 = 0$$
$$\Rightarrow (\alpha - \beta + 4)(\alpha - \beta) = 0$$
$$\because \alpha \neq \beta$$
$$\therefore \alpha - \beta + 4 = 0$$
$$\alpha - \beta = -4$$
$$\therefore |\alpha - \beta| = 4$$

Q39. Let in a series of $2n$ observations, half of them are equal to a and remaining half are equal to $-a$. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20 respectively. Then the value of $a^2 + b^2$ is equal to :

(A) 425 (B) 925 (C) 650 (D) 250

Ans. A

Sol. Let observations are denoted by x_i

For $1 \leq i < 2n$

$$\bar{x} = \frac{\sum x_i}{2n} = \frac{(a + a + \dots + a) - (a + a + \dots + a)}{2n}$$

$$\bar{x} = 0$$

$$\text{and } (\sigma_x)^2 = \frac{\sum (x_i)^2}{2n} - (\bar{x})^2 = \frac{a^2 + a^2 + \dots + a^2}{2n} - 0 = a^2$$

$$\Rightarrow \sigma_x = a$$

Now, adding a constant 'b' then,

$$\bar{y} = \bar{x} + b = 5]$$

$$\Rightarrow b = 5$$

and $\sigma_y = \sigma_x$ (No change in S.D.)

$$\Rightarrow a = 20$$

$$\Rightarrow a^2 + b^2 = 425$$

Q40.

Ans. -

Sol. -

Q41. Consider a set of $3n$ numbers having variance 4. In this set, the mean of first $2n$ numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first $2n$ numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k , then $9k$ is equal to.

Ans. 68

Sol. First $3n$ number $\rightarrow a_1, a_2, a_3, \dots, a_{3n}$

First n number $\rightarrow b_1, b_2, \dots, b_n$

$$\sigma^2 = \sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} = (\bar{x})^2 \quad \bar{x} = \frac{2n \times 6 + n \times 3}{3n}$$

$$\sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} - 5^2 \quad \bar{x} = 5$$

$$\mu = \frac{\sum a^2 + \sum b^2}{3n} - 25$$

$$\sum a^2 + \sum b^2 = 87n$$

Now first $2n$ numbers $\rightarrow a_1 + 1, a_2 + 1, \dots, a_{2n} + 1$

Last n numbers $b_1 - 1, b_2 - 1, \dots, b_n - 1$

$$\sigma^2 = \frac{\sum (a+1)^2 + \sum (b-1)^2}{3n} - (\bar{x})^2$$

$$\sigma^2 = \sum a^2 + 2n + 2 \sum a +$$

$$\sigma^2 = \frac{\left(\sum a^2 + 2n + 2 \sum a \right) + \left(\sum b^2 + n - 2 \sum b \right)}{3n} - \left(\frac{16}{3} \right)^2$$

$$\sigma^2 = \frac{87n + 3n + 2(12n) - 2(3n)}{3n} - \left(\frac{16}{3} \right)^2$$

$$\sigma^2 = \left(\frac{108}{3}\right) \left(\frac{16}{3}\right)^3$$

$$\sigma^2 = \frac{68}{9}$$

$$\text{then Ans. } 9k = 9 \left(\frac{68}{9}\right) = 68$$

Q42. Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to :

Ans. 5

Sol. For observation I :

$$\sum x_i = 10 \times 2 = 20$$

$$\sigma^2 = \frac{\sum x_i^2}{10} - (2)^2 = 2$$

$$\sum x_i^2 = 60$$

For observation II :

$$\sum x_i = n \times 3 = 3n$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - (3)^2 = 1$$

$$\sum x_i^2 = 10n$$

For combined (n + 10) observations

$$\sum x_i = 20 + 3n$$

$$\sum x_i^2 = 60 + 10n$$

$$\sigma^2 = \left(\frac{60 + 10n}{n + 10}\right) - \left(\frac{20 + 3n}{n + 10}\right)^2 = \frac{17}{9}$$

$$n = 5$$

Q43. Consider three observations a, b and c such that $b = a + c$. If standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is **true** ?

$$(A) b^2 = a^2 + c^2 + 3d^2 \quad (B) 2 = 3(a^2 + c^2) - 9d^2 \quad (C) b^2 = 3(a^2 + c^2) + 9d^2$$

$$(D) b^2 = 3(a^2 + c^2 + d^2)$$

Ans. B

Sol. $AM = \bar{x} = \frac{a + b + c}{3} = \frac{2b}{3}$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$d^2 = \left(\frac{a^2 + b^2 + c^2}{3}\right) - \left(\frac{2b}{3}\right)^2$$

$$9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$$

$$b^2 = 3a^2 + 3c^2 - 9d^2$$

Q44. The mean of 6 distinct observations is 6.5 and their variance is 10.25. If 4 out of 6 observations are 2, 4, 5 and 7, then the remaining two observations are :

(A) 3, 18 (B) 1, 20 (C) 10, 11 (D) 8, 13

Ans. C

Sol. $\bar{x} = 6.5$

$$\frac{18 + x + y}{6} = 6.5$$

$$\Rightarrow x + y = 21 \quad \dots\dots\dots(i)$$

$$\sigma^2 = 10.25$$

$$\frac{\sum x^2}{n} - (\bar{x})^2 = 10.25$$

$$\frac{94 + x^2 + y^2}{6} - (6.5)^2 = 10.25$$

$$x^2 + y^2 = 221$$

$$(x + y)^2 - 2xy = 221$$

$$xy = 110 \quad \dots\dots\dots(ii)$$

solving equation (i) & (ii)

$$x = 10, y = 11$$

Q45. If the mean and variance of six observations 7, 10, 11, 15, a, b are and $\frac{20}{3}$, respectively, then the value of $|a - b|$ is equal to :

(A) 11 (B) 1 (C) 7 (D) 9

Ans. B

Sol. $\text{mean} = \frac{7 + 10 + 11 + 15 + a + b}{6} = 10$

$$4 + b = 17$$

$$\text{variance} = \frac{7^2 + 10^2 + 11^2 + 15^2 + a^2 + b^2}{6} - = 100 \frac{20}{3}$$

$$a^2 + b^2 = 145$$

$$(a + b)^2 - 2ab = 145$$

$$ab = 72$$

$$(a - b)^2 = (a + b)^2 - 4ab$$

$$|a - b| = 1$$

- Q46. The first of the two samples in a group has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then the standard deviation of the second sample is :

(A) 6 (B) 4 (C) 5 (D) 8

Ans. B

Sol. $15.6 = \frac{100 \times 15 + 150 \times \bar{x}}{250}$

$$\bar{x} = 16$$

$$\text{Combined standard deviation} = \sqrt{13.44}$$

$$\Rightarrow \text{Combined variance } (\sigma^2) = 13.44$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$13.44 = \frac{\sum x_i^2}{250} - 243.36$$

$$\Rightarrow \sum x_i^2 = 64200$$

For 1st sample

$$9 = \frac{\sum x_I^2}{100} - 225$$

$$\Rightarrow \sum x_I^2 = 23400$$

For 2nd sample

$$\sum x_{II}^2 = 64200 - 23400 = 40800$$

standard deviation of IInd sample will be

$$\sqrt{\frac{\sum x_{II}^2}{n} - (\bar{x}_{II})^2} = \sqrt{\frac{40800}{150} - 256} = 4$$

- Q47. If the mean and variance of the following data :

6, 10, 7, 13, a, 12, b, 12

are 9 and $\frac{37}{4}$ respectively, then $(a - b)^2$ is equal to :

(A) 12 (B) 24 (C) 16 (D) 32

Ans. C

Sol. $\bar{x} = \frac{6 + 10 + 7 + 13 + a + 12 + b + 12}{8} = 9$

$$a + b = 12$$

$$\sigma^2 = \frac{\sum xi^2}{8} - (\bar{x})^2 = \frac{37}{4}$$

$$\Rightarrow \frac{6^2 + 10^2 + 7^2 + 13^2 + a^2 + 12^2 + b^2 + 12^2}{8} - 81 = \frac{37}{4}$$

$$\Rightarrow a^2 + b^2 = 80$$

$$(a - b)^2 + (a + b)^2 = 2(a^2 + b^2)$$

$$(a - b)^2 = 160 - 144 = 16$$

Q48. Let the mean and variance of the frequency distribution

x	x ₁ =	x ₂ =	x ₃ =	x ₄ =
	2	6	8	9
f	4	4	α	β

be 6 and 6.8 respectively. If x₃ is changed from 8 to 7, then the mean for the new data will be:

(A) 5 (B) 4 (C) $\frac{16}{3}$ (D) $\frac{17}{3}$

Ans. D

Sol. $\mu = \frac{\sum f_i x_i}{\sum f_i}$

$$\Rightarrow \mu = \frac{2 \times 4 + 6 \times 4 + 8 \times \alpha + 9 \times \beta}{4 + 4\alpha + \beta}$$

$$\Rightarrow 6 = \frac{32 + 8\alpha + 9\beta}{8 + \alpha + \beta}$$

$$\Rightarrow 48 + 6\alpha + 6\beta = 32 + 8\alpha + 9\beta$$

$$\Rightarrow 2\alpha + 8\beta = 16$$

$$\sigma^2 = \frac{\sum f_i \cdot (x_i - \mu)^2}{\sum f_i}$$

$$\Rightarrow 6.8 = \frac{4 \times 16 + 4 \times 0 + \alpha \times 4 + \beta \times 9}{8 + \alpha + \beta}$$

$$\Rightarrow \frac{34}{5} = \frac{4 + 4\alpha + 9\beta}{8 + \alpha + \beta}$$

$$\Rightarrow 14\alpha - 11\beta = 48$$

On solving (i) & (ii)

$$\alpha = 5 \text{ and } \beta = 2$$

$$\begin{aligned} \text{New mean } (\mu) &= \frac{4 \times 2 + 4 \times 6 + \alpha \times 7 + \beta \times 9}{8 + \alpha + \beta} \\ &= \frac{8 + 24 + 5 \times 7 + 2 \times 9}{8 + 5 + 2} \\ &= \frac{85}{15} = \frac{17}{3} \end{aligned}$$

Q49. Consider the following frequency distribution :

Class :	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then the value $(a - b)^2$ is equal to.

Ans. 4

Sol.

Class	x_i	f_i	$f_i x_i$
0 – 6	3	a	3a
6 – 12	9	b	9b
12 – 18	15	12	180
18 – 24	21	9	189
24 – 30	27	5	135

$$\text{Mean} = \frac{3a + 9b + 180 + 189 + 135}{a + b + 12 + 9 + 5} = \frac{309}{22}$$

$$18a + 37b = 1018 \quad \dots\dots(1)$$

$$\text{Now median} = 12 + \left(\frac{\frac{a+b+26}{2} - a - b}{12} \right) \times 6 = 14$$

$$a + b = 18 \quad \dots\dots(2)$$

by (1) & (2)

$$a = 8, \quad b = 10$$

$$(a - b)^2 = 4$$

Q50. Consider the following frequency distribution

Class	10-20	20-30	30-40	40-50	50-60
Frequency	α	110	54	30	β

If the sum of all frequency is 584 and median is 45, then $|\alpha - \beta|$ is equal to

(A) 132 (B) 164 (C) 186 (D) 198

Ans. B

Sol.

Class	Frequency	c.f.
10-20	α	α
20-30	110	$\alpha + 110$

30-40	54	$\alpha + 164$
40-50	30	$\alpha + 194$
50-60	β	$\alpha + \beta + 194$
	$N = 584$	

Given, sum of frequency = 584 $\Rightarrow \alpha + \beta = 390$

$$\text{Median} = l + \left[\frac{\left(\frac{N}{2}\right) - c.f.}{f} \right] \times n$$

$$Q \frac{N}{2} = \frac{584}{2} = 292 \quad \therefore M = 40 + \left[\frac{292 - (\alpha + 164)}{30} \right] \times 10$$

$$\Rightarrow 45 = 40 + \left(\frac{128 - \alpha}{3} \right)$$

$$\Rightarrow 15 = 128 - \alpha \Rightarrow \alpha = 113 \text{ and } \beta = 277$$

$$\therefore |\alpha - \beta| = |113 - 277| = 164$$

Q51. The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is :

- (A) (11, 25) (B) (10.5, 25) (C) (11, 26) (D) (10.5, 26)

Ans. D

Sol. $n = 20$ $\bar{x} = 10$ $\sigma = 2.5$

$$\Sigma \bar{x} = n\bar{x} = 200$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

$$6.25 = \frac{\Sigma x_i^2}{20} - 100$$

$$\frac{\Sigma x_i^2}{20} = 106.25$$

$$\Sigma x_i^2 = 2125$$

For new distribution

$$\Sigma x_i = 200 - 25 + 35 = 210 \Rightarrow \bar{x} = \frac{\Sigma x_i}{20} = \frac{210}{20} = 10.5$$

$$\Sigma x_i^2 = 2125 - (25)^2 + (35)^2 = 2725$$

$$\begin{aligned} \beta = \sigma^2 &= \frac{\Sigma x_i^2}{20} - (\bar{x})^2 = \frac{2725}{20} - (10.5)^2 \\ &= 136.25 - 110.25 \\ &= 26 \end{aligned}$$

$$(\alpha, \beta) \equiv (10.5, 26)$$

Q52. The mean and variance of 7 observations are 8 and 16 respectively. If two observations are 6 and 8, then the variance of the remaining 5 observations is :

- (A) $\frac{134}{5}$ (B) $\frac{112}{5}$ (C) $\frac{536}{25}$ (D) $\frac{92}{5}$

Ans. C

Sol. $x_1, x_2, x_3, x_4, x_5, 8, 6$

$$\frac{x_1 + x_2 + x_3 + x_4 + x_5 + 14}{7} = 8$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 42 \quad \dots\dots\dots (1)$$

$$\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + 8^2 + 6^2}{7} - 64 = 16$$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 = 460$$

So variance of $x_1, x_2, \dots, x_5$

$$= \frac{460}{5} - \left(\frac{42}{5}\right)^2 = \frac{536}{25}$$

Q53. Let the mean and variance of the four numbers 3, 7, x and y ($x > y$) be 5 and 10 respectively. Then the mean of four numbers $3 + 2x, 7 + 2y, x + y$ and $x - y$ is _____.

Ans. 12

Sol. -

Q54. Let n be an odd natural number such that the variance of 1, 2, 3, 4, ..., n is 14. Then n is equal to _____.

Ans. 13

Sol. $\sigma^2 = \left(\frac{\sum n^2}{n}\right) - \left(\frac{\sum n}{n}\right)^2 = 14$
 $\Rightarrow n^2 = 169 \Rightarrow n = 13$

Q55. An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to _____.

Ans. 7

Sol. $\sigma_b^2 = 2$ $n_1 = \text{No of boys} = 20$
 $\bar{x}_b = 12$ $n_2 = \text{No. of girls} = 30$
 $\sigma_g^2 = 2$
 $\bar{x}_g =$

$$\bar{x} = 15$$

$$\text{Now } \bar{x} = \frac{\bar{x}_b n_1 + \bar{x}_g n_2}{n_1 + n_2}$$

$$15 = \frac{12 \times 20 + \bar{x}_g \times 30}{30 + 20}$$

$$750 - 240 = 30 \bar{x}_g$$

$$\bar{x}_g = 17$$

$$\mu = 17$$

variance of combined group

$$\sigma^2 = \frac{n_1 \sigma_b^2 + n_2 \sigma_g^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x}_b - \bar{x}_g)^2$$

$$\sigma^2 = \frac{20 \times 2 + 30 \times 2}{20 + 30} + \frac{20 \times 30}{(20 + 30)^2} (12 - 17)^2$$

$$\sigma^2 = 8$$

$$\Rightarrow \mu + \sigma^2 = 17 + 8$$

$$= 25$$

Q56. The mean of 10 numbers $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$ is _____.

Ans. 398

$$\text{Sol. } T_r = (3r + 4) \cdot (2r + 6)$$

$$T_r = 6r^2 + 26r + 24$$

$$\bar{x} = \frac{\sum_{r=1}^{10} T_r}{10} = \frac{6 \cdot \frac{10 \times 11 \times 21}{6} + 26 \cdot \frac{10 \times 11}{2} + 24 \times 10}{10}$$

$$\bar{x} = 398$$

Q57. Let X be a random variable with distribution.

x	-2	-1	3	4	6
P(X=x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2 , then $\frac{100\sigma^2}{781}$ is equal to :

Ans. 1

$$\text{Sol. } \sum P_i = 1$$

$$\Rightarrow \frac{1}{5} + a + \frac{1}{3} + \frac{1}{5} + b = 1$$

$$\Rightarrow a + b = 1 - \left(\frac{2}{5} + \frac{1}{3} \right)$$

$$\Rightarrow a + b = 1 - \frac{11}{15}$$

$$\therefore a + b = \frac{4}{15} \quad \dots\dots\dots(1)$$

$$\mu = \sum p_i x_i$$

$$\Rightarrow 2.3 = \frac{-2}{5} - a + 1 + \frac{4}{5} + 6b$$

$$a - 6b = 1 + \frac{2}{5} - \frac{23}{10}$$

$$a - 6b = \frac{10 + 4 - 23}{10}$$

$$a - 6b = \frac{-9}{10} \quad \dots\dots\dots(2)$$

Solving the equatino (1) and (2)

$$a = \frac{1}{10}, b = \frac{1}{6}$$

using the formula $\sigma^2 = \Sigma p_i x_i^2 - (\bar{x})^2$

$$= \frac{1}{5} \times 4 + \frac{1}{10} \cdot (1) + \frac{1}{3} \times 9 + \frac{1}{5} \times 16 + \frac{1}{6} \times 36$$

$$- (2.3)^2$$

$$= \frac{4}{5} + \frac{1}{10} + 9 + \frac{16}{5} - \left(\frac{23}{10}\right)^2$$

$$\sigma^2 = 13 + \frac{1}{10} - \frac{529}{100} = \frac{781}{100}$$

$$100 \sigma^2 = 781$$

Q58. The mean of the number a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to :

- (A) 60 (B) 55 (C) 50 (D) 45

Ans. A

Sol.

$$\sigma^2 = \frac{\sum_{i=1}^5 (x_i - \bar{x})^2}{5}$$

Mean = 6

$$\frac{a + b + 8 + 5 + 10}{5} = 6$$

$$a + b = 7$$

$$b = 7 - a$$

$$6.8 = \frac{(a - 6)^2 + (b - 6)^2 + (8 - 6)^2 + (5 - 6)^2 + (10 - 6)^2}{5}$$

$$34 = (a - 6)^2 + (7 - a - 6)^2 + 4 + 1 + 16$$

$$a^2 - 7a + 12 = 0 \Rightarrow a = 4 \text{ or } a = 3$$

$$a = 4 \quad a = 3$$

$$b = 3 \quad b = 4$$

$$M = \frac{\sum_{i=1}^5 |x_i - \bar{x}|}{5}$$

$$M = \frac{|a - 6| + |b - 6| + |8 - 6| + |5 - 6| + |10 - 6|}{5}$$

when a = 3, b = 4

when a = 4, b = 3

$$M = \frac{3 + 2 + 2 + 1 + 4}{5}$$

$$M = \frac{2 + 3 + 2 + 1 + 4}{5}$$

$$M = \frac{12}{5} \qquad M = \frac{12}{5}$$

$$25M = 25 \times \frac{12}{5} = 60$$

Q59. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :

(A) 10 (B) 36 (C) 43 (D) 60

Ans. C

Sol. Given $\bar{x} = 15$

$$\sigma = 2 \Rightarrow \sigma^2 = 4$$

$$x_1 + x_2 + \dots + x_{50} = 15 \times 50 = 750$$

$$4 = \frac{x_1^2 + x_2^2 + \dots + x_{50}^2}{50} - 225$$

$$x_1^2 + x_2^2 + \dots + x_{50}^2 = 50 \times 229$$

Let a be the correct observation and b is the incorrect observation

$$a + b = 70$$

$$16 = \frac{750 - b + a}{50}$$

$$a - b = 50$$

$$= a = 60 \quad b = 10$$

$$\text{correct variance} = \frac{50 \times 229 + (60)^2 - (10)^2}{50} - 256$$

$$= 43$$

Let a be the correct observation and b is the incorrect observation

$$a + b = 70$$

Q60. The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y, where $x < y$, are 6 and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to :

(A) 162 (B) 320 (C) 674 (D) 420

Ans. B

$$\text{Sol. } \frac{36 + x + y}{8} = 6 \Rightarrow \frac{9}{4} = \frac{16 + 25 + 36 + 36 + 49 + 64 + x^2 + y^2}{8} - (6)^2$$

$$x + y = 12 \Rightarrow x^2 + y^2 = 306 - 226$$

$$\Rightarrow x^2 + y^2 = 80$$

$$x = 4, y = 8$$

$$x^4 + y^2 = 256 + 64 \Rightarrow 320$$

- Q61. Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observation are $\frac{7}{2}$ and a respectively, the $(4a + x_5)$ is equal to :
- (A) 13 (B) 15 (C) 17 (D) 18

Ans. B

Sol. Mean = $\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{24}{5}$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 + x_5 = 24$$

$$\Rightarrow \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 14$$

$$\frac{\sum_{i=1}^5 x_i^2}{5} - \left(\frac{24}{5}\right)^2 = \frac{194}{25}$$

$$\sum_{i=1}^5 x_i^2 = 154 \Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 = 154 - 100 = 54$$

$$\Rightarrow x_5 = 10$$

$$\frac{\sum_{i=1}^4 x_i^2}{4} - \left(\frac{7}{2}\right)^2 = a$$

$$\Rightarrow a = \frac{5}{4}$$

$$x_5 + 4a = 10 + 5 = 15$$

- Q62. The number of values of $a \in \mathbb{N}$ such that the variance of 3, 7, 12, a , $43 - a$ is a natural number is :
- (A) 0 (B) 2 (C) 5 (D) infinite

Ans. A

Sol. Given numbers : 3, 7, 12, a, 43 - a

$$\text{Mean} = \bar{x} = \frac{3 + 7 + 12 + a + 43 - a}{5} = 13$$

$$\text{Variance} = \frac{9 + 49 + 144 + a^2 + (43 - a)^2}{5} - 13^2$$

So, variance $\in$ natural number

$$\frac{2a^2 - 86a + 1}{5} \in N$$

$$2a^2 - 86a + 1 = 5N$$

$$2a^2 - 86a + 1 - 5N = 0$$

It is quadratic in a

So, Δ cannot be perfect square

So, a cannot be perfect square

$\therefore$ Number of values = 0

Q63. If the mean deviation about the mean of the numbers 1, 2, 3,, n where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.

Ans. 21

Sol.
$$\bar{x} = \frac{1 + 2 + 3 + \dots + n}{n} = \frac{n+1}{2}$$

$$\sum |x_i - \bar{x}| = |1 - \bar{x}| + |2 - \bar{x}| + \dots + \left| \frac{n+1}{2} - \bar{x} \right| + \dots + |n - \bar{x}|$$

$$= 2 \cdot \left\{ |1 - \bar{x}| + |2 - \bar{x}| + \dots + \left| \frac{n-1}{2} - \bar{x} \right| \right\}$$

$$= 2 \left\{ |\bar{x} - 1| + |\bar{x} - 2| + \dots + \left(\bar{x} - \frac{n-1}{2} \right) \right\}$$

$$= 2 \left\{ \frac{n-1}{2} \cdot \bar{x} - \frac{\left(\frac{n-1}{2} \cdot \frac{n+1}{2} \right)}{2} \right\}$$

$$= \frac{(n-1) \cdot (n+1)}{2} - \frac{(n-1) \cdot (n+1)}{4}$$

$$\sum |x_i - \bar{x}| = \frac{n^2 - 1}{4}$$

$$\frac{\sum |x_i - \bar{x}|}{n} = \frac{n^2 - 1}{4n} = \frac{5n + 5}{n}$$

$$\Rightarrow n^2 - 20n - 21 = 0$$

$$(n+1)(n-21) = 0$$

$$n = 21$$

Q64. The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

Ans. 17

Sol. We have

$$\text{Variance} = \frac{\sum_{r=1}^{15} X_r^2}{15} - \left(\frac{\sum_{r=1}^{15} X_r}{15} \right)^2$$

Now, as per information given in equation $\frac{\sum X_r^2}{15} - 8^2 = 3^2 \Rightarrow \sum X_r^2 = 1095$

Now, the new $\sum X_r = (15 \times 8) - 5 + (20) = 135$

$$\therefore \text{Variance} = \frac{1470}{15} - \left(\frac{135}{15} \right)^2 = 98 - 81 = 17$$

Q65. Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62 and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is _____.

Ans. 0

Sol. According to given data

$$\bar{x} = 62, \frac{\sum (x_i - \bar{x})^2}{7} = 20$$

$$\Rightarrow (x_1 - 62)^2 + (x_2 - 62)^2 + \dots \dots \dots (x_7 - 62)^2 = 140 \text{ for failure of any student } x_i < 50$$

Suppose $x_1 = 49$.

$(49 - 62)^2 = 169$ which is not possible. Therefore no student is going to score less than 50.

= 0 students

Q66. If the mean deviation about median for the numbers 3, 5, 7, 2k, 12, 16, 21, 24, arranged in the ascending order, is 6 then the median is :

(A) 11.5 (B) 10.5 (C) 12 (D) 11

Ans. D

Sol. Median = $\frac{2k + 12}{2} = k + 6$

$$M.D. = \frac{\sum |x_i - M|}{n} = 6$$

$$= \frac{(k + 3) + (k + 1) + (k - 1) + (6 - k) + (6 - k) + (10 - k) + (15 - k) + (18 - k)}{8}$$

$$\Rightarrow \frac{58 - 2k}{8} = 6$$

$$\Rightarrow k = 5$$

$$\text{Median} = 5 + 6 = 11$$

Q67. The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $P(X = 1) = \frac{4}{243}$, then $P(X = 4 \text{ or } 5)$ is equal to :

- (A) $\frac{5}{9}$ (B) $\frac{64}{81}$ (C) $\frac{16}{27}$ (D) $\frac{145}{243}$

Ans. C

Sol. $np = \alpha$ (1)

$npq = \alpha/3$ (2)

From (1) & (2)

$q = 1/3$ & $p = 2/3$

$${}^nC_1 q^{n-1} p^1 = \frac{4}{243}$$

$$\frac{n}{3^n} = \frac{2}{243}$$

$n = 6$

$$P(4 \text{ or } 5) = {}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^6C_5 \left(\frac{2}{3}\right)^5 \cdot \left(\frac{1}{3}\right)^0$$

$$= \frac{16}{27}$$

Q68. The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Ans. 238

Sol. Wrong mean $= \mu_1 = 30$

Wrong S.D $= \sigma_1 = 5$

$$\frac{\sum x_i}{40} = 30$$

$$\Rightarrow \sum x_i = 1200$$

$$\sigma_1^2 = 25$$

$$\Rightarrow \frac{\sum x_i^2}{40} - 30^2 = 25$$

$$\Rightarrow \sum x_i^2 = 925 \times 40 = 37000$$

New sum $= \sum x_i' = 1200 - 10 - 12 = 1178$

New mean $= \mu_i' = \frac{1178}{38} = 31$

$$\text{New } \Sigma x_i^2 = 37000 - (10)^2 - (12)^2 = 36756$$

$$\text{New S.D, } \sigma'_1 = \sqrt{\frac{36756}{38} - (31)^2} = \sigma$$

$$36756 - (31)^2 \times 38 = 38\sigma^2$$

$$\Rightarrow 38\sigma^2 = 238$$

Q69. The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is _____.

Ans. 2

$$\text{Sol. } n = 10, \bar{x} = \frac{\sum x_i}{10} = 15$$

$$6^2 = \frac{\sum x_i^2}{10} - (\bar{x})^2 = 15$$

$$\Rightarrow \sum_{i=1}^{10} x_i = 150$$

$$\Rightarrow \sum_{i=1}^9 x_i + 25 = 150$$

$$\Rightarrow \sum_{i=1}^9 x_i = 125$$

$$\Rightarrow \sum_{i=1}^9 x_i + 15 = 140$$

$$\text{Actual mean} = \frac{140}{10} = 14 = \bar{x}_{new}$$

$$\sum_{i=1}^9 \frac{x_i^2 + 25^2 - 15^2}{10} = 15$$

$$\Rightarrow \sum_{i=1}^9 x_i^2 + 625 = 2400$$

$$\sum_{i=1}^9 x_i^2 = 1775$$

$$\sum_{i=1}^9 x_i^2 + 15^2 = 2000 = (\Sigma x_i^2)_{actual}$$

$$6_{actual}^2 = \frac{(\Sigma x_i^2)_{actual} - (\bar{x}_{new})^2}{10}$$

$$= \frac{2000}{10} - 14^2$$

$$= 200 - 196 = 4$$

$$(\text{S.D})_{actual} = 6 = 2$$

Q70. Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal $\bar{x}$ to is _____.

Ans. 142

Sol.

$$\sum x_0^1 = \frac{3 \left(1 - \left(\frac{1}{2}\right)^{20}\right)}{1 - \frac{1}{2}} = 6 \left(1 - \frac{1}{2^{20}}\right)$$

$$= \sum_{i=1}^{20} (x_{i+1})^2$$

$$= \sum_{i=1}^{20} (x_i)^2 + (i) - 2x_i i$$

$$\text{Now} = \sum_{i=1}^{20} (x_i)^2 = \frac{9 \left(1 - \left(\frac{1}{4}\right)^{20}\right)}{1 - \frac{1}{4}} = 12 \left(1 - \frac{1}{2^{40}}\right)$$

$$\sum_{i=1}^{20} i^2 = \frac{1}{6} \times 20 \times 21 + 41 = 2870$$

$$\sum_{i=1}^{20} x_i i = s = 3 + 2.3 \frac{1}{2} + 3.3 \frac{1}{2^2} + 4.3 \frac{1}{2^3} + \dots \text{AGP}$$

$$= 6 \left(-\frac{22}{2^{20}}\right)$$

$$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12 \left(2 - \frac{22}{2^{20}}\right)}{20}$$

$$\bar{x} = \frac{2858}{20} + \left(\frac{-12}{2^{40}} + \frac{22}{2^{20}}\right) \times \frac{1}{20}$$

$$[\bar{x}] = 142$$

Q71. Let the mean and the variance of 20 observations $x_1, x_2, \dots, x_{20}$ be 15 and 9, respectively. For $\alpha \in \mathbb{R}$, if the mean of $(x_1 + \alpha)^2, (x_2 + \alpha)^2, \dots, (x_{20} + \alpha)^2$ is 178, then the square of the maximum value of α is equal to _____.

Ans. 4

Sol. $\sum x_1 = 15 \times 20 = 300 \quad \dots(i)$

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \quad \dots(ii)$$

$$\sum x_1^2 = 234 \times 20 = 4680$$

$$\frac{\sum (x_1 + \alpha)^2}{20} = 178 \Rightarrow \sum (x_1 + \alpha)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2\alpha \sum x_1 + \sum \alpha^2 = 3560$$

$$4680 + 600\alpha + 20\alpha^2 = 3560$$

$$\Rightarrow \alpha^2 + 30\alpha + 56 = 0$$

$$\Rightarrow (\alpha + 28)(\alpha + 2) = 0$$

$$\alpha = -2, -28$$

Square of maximum value of α is 4.

Q72. The sum and product of the mean and variance of a binomial distribution are 82.5 and 1350 respectively. Then the number of trials in the binomial distributions is _____.

Ans. 96

Sol. Let, mean = $m = np$

& variance = $v = npq$, $p + q = 1$

$$\text{Sum} = m + v = \frac{165}{2}$$

$$\text{Product} = mv = 1350$$

On solving,

$$m = np = 60 \text{ \& } v = npq = \frac{45}{2} \therefore q = \frac{3}{8} \therefore P = \frac{5}{8}$$

Hence $n = 96$